

SHENG ZHAO

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My research interests include interactive **usability**, **privacy** and **security** in **Virtual Reality**, as well as efficient and trustworthy **human-AI interaction** technologies.

EDUCATION

Tsinghua University, Beijing, China 2021.09 – Present
B.Sc. in **Mathematics and Physics** & B.Eng. in **Electrical Engineering** Dual degree

- Junior, GPA: 3.81/4.
- A+: Optics, Quantum Mechanics; A: Electromagnetism, Thermodynamics, Calculus, Mathematical Physics Equations, etc.

PUBLICATIONS & MANUSCRIPTS

- **Sheng Zhao**, JunRay Zhu, Xueyang Wang, Hongyi Li, Fang Yi, Hewu Li[†], Shuning Zhang[†] and Xin Yi[†]. CoordAuth: A Two-factor Authentication Method in Virtual Reality Leveraging Head-Eye Coordination. *Ubicomp'24*, under review. [PDF](#).
- Xueyang Wang*, **Sheng Zhao***, Yihe Wang, Ziyu Han, Xin Tong[†] and Xin Yi[†]. EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in Virtual Reality. *ISMAR'24*, in submission.

RESEARCH EXPERIENCES

EmoBlend: Revealing the Impact of Facial Blendshape Intensity on Emotional Perception in VR 2023.04 – 2024.01
Research Assistant, Advisor: *Xin Yi, Yuanchun Shi* Tsinghua University

- Implementing VR face2face chat using TCP-based socket and Oculus-Movement sdk.
- Designing both qualitative and quantitative research methods based on interviews and questionnaires, conducting user studies and data analysis.
- Proposing an improved camera2blendshape method to mitigate the uncanny valley effect and enhance the emotional experience.
- In submission to *ISMAR'24*. [Details](#).

CoordAuth: A Two-factor Authentication Method in VR Leveraging Head-Eye Coordination 2023.10 – 2024.01
Research Assistant, Advisor: *Xin Yi* Tsinghua University

- Project leader, proposing the current two-factor authentication framework based on unlock pattern and behavior biometrics.
- Using Unity to build the 3x3 unlock UI leveraging gaze interaction in VR.
- Designing the biometric classifier by machine learning, using RF, etc. to implement the classification of behavior biometrics during saccades and fixations of eye movements.
- Conducting user studies, validating framework robustness against password collision, contextual factors, and shoulder surfing attacks.
- Submitted to *Ubicomp'24*. [PDF](#). [Details](#).

Ongoing research experiences can be found [here](#).

TECHNOLOGY COMPETITIONS

Tsinghua University Software Design Competition 2022.10 – 2023.01
Developed a Smart Home app using Java, Implemented weather forecasting and control of indoor device operation status through Bluetooth. [Code](#).

The 12th Tsinghua University Digital System Design Competition 2022.08 – 2022.10
Designed a small car driven by MSP430 with various sensors for autonomous navigation and resource acquisition. [Details](#).

SKILLS

- Program languages: C/C++/C#, Python, Java, MATLAB and Verilog.
- Tools and Frameworks: Git, LaTeX, Unity, PyTorch, Scikit-learn, Markdown and MSP430.
- Techniques: VR development, Quantitative and Qualitative research.

SELECTED AWARDS

- *First Prize*, 36th China Physics Olympiad (CPHO), 0.9% 2020
- *First Prize*, The 12th Tsinghua University Digital System Design Competition, 1.7% 2022
- *Second Prize*, Tsinghua University "Smart Coastal" Software Design Competition 2023
- *First Prize*, Tsinghua University Technological Innovation Scholarship 2023
- *First Prize*, The 3rd Tsinghua Craftsman Competition, 1%, Presented a speech 2023